

DIPOLE TYPE BY LENGTH

Compute first $\lambda=c/f$, then ratio l/λ		
l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\ \Omega$
$= 1$	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\ \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$
$S(R, \theta)$ — avg. pwr dens. at (R, θ) (W/m ²)
l — antenna length (m, the rod itself)
I_0 — peak current in dipole (A)
R — distance from antenna to obs. point (m)
θ — angle from dipole axis (rad/°)
$k=2\pi/\lambda$ — wavenumber (rad/m)
$\eta_0=120\pi \approx 377\ \Omega$ — free-space impedance
Max direction broadside $\theta_{\text{max}}=\pi/2$
Total radiated power integrate over 4π
$P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda} \right)^2 \text{ (W)}$
Directivity exact $D_{\text{Hertz}}=1.5$ (1.76 dB)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l
$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$
At $\theta=0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S=0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta=0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.
Normalized pattern dimensionless $F(\theta)=S(\theta)/S_{\text{max}}$
Quarter-wave $l=\lambda/4$ $\pi l/\lambda=\pi/4$, $\cos(\pi/4)=\sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$, $S_0 = \frac{15I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$
Half-wave $l=\lambda/2$ $\pi l/\lambda=\pi/2$, shortcut form
$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta}$
$D=1.64$, $R_{\text{rad}}=73\ \Omega$, $\theta_{\text{max}}=\pi/2$.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

°	rad	$\sin \theta$	$\cos \theta$	$\sin^2 \theta$	$\cos^2 \theta$	θ
0	0	0	1	0	1	
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4	
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2	
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4	
90	$\pi/2$	1	0	1	0	
180	π	0	-1	0	1	

Convert: $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \pi/180$. Antenna formulas usually take θ in rad.

BEAM PARAMETERS — β, Ω_p, D

Step 1 — Normalize always start here
$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)
Step 2 — HPBW β half-power beamwidth, full angular width at -3 dB
Set $F(\theta)=0.5$, solve for $\theta \rightarrow \theta_{1/2}$
$\beta = 2\theta_{1/2}$ (symmetric beams, rad or °)
Gaussian:
$F(\theta_{1/2}) = e^{-a\theta^2/2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$
Step 3 — Pattern solid angle Ω_p units: sr
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$
No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.
TI-36X Pro: [2nd] [d/dx] for $\int_0^\pi$, set RAD mode, multiply result by 2π .
Narrow-beam Gaussian shortcut (sin $\theta \approx \theta$, $\rightarrow \infty$):
$\Omega_p \approx \frac{\pi}{a} = \frac{\pi \theta_{1/2}^2}{\ln 2}$ ($F=e^{-a\theta^2/2}$)
Step 4 — Directivity D
$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$
Alt. with two orthogonal HPBW's: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$.
Step 5 — Gain (if asked) $\xi = \text{rad. eff.}$, $0 < \xi \leq 1$
$G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G=D$.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	$R_{\text{rad}} \text{ (}\Omega\text{)}$
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
$\lambda/4$	$\lambda/4$ (gnd)	1.64	2.15	36.6
monopole				
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}}=36.6I_0^2 W$. $\lambda/4$ monopole (above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.
Lossless antenna ($\xi=1$) $\Rightarrow G=D$. So in Friis, use $G_t, G_r =$ these D values (e.g. half-wave $G=1.64$).

EFFECTIVE AREA A_e

Definition antenna's RX cross section (m²)
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)
Physical cross section wire dipole, diameter d
$A_p = l d = \frac{\lambda}{2} d$ (half-wave)
Inverse: D from A_e
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)
Hertzian special case $D=1.5$
$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119\lambda^2 \text{ (m}^2\text{)}$
Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

R_{rad} , EFFICIENCY, POWER SPLIT

Radiation resistance Ω , equiv. ckt. load absorbing P_{rad}
$R_{\text{rad}} = \frac{2P_{\text{rad}}}{I_0^2}$ $R_{\text{loss}} = \frac{2P_{\text{loss}}}{I_0^2}$
Input power split P_t from generator
$P_t = P_{\text{rad}} + P_{\text{loss}}$ $R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}$
Radiation efficiency two equivalent forms
$\xi = \frac{P_{\text{rad}}}{P_t} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}$ (unitless)
Receiving antenna $\vec{V}_{\text{oc}} = \vec{E}^i l$ for short dipole
Max pwr transfer when $Z_L = Z_{\text{in}}^*$ (conjugate match):
$P_L = P_{\text{int}} = \frac{ \vec{V}_{\text{oc}} ^2}{8R_{\text{rad}}} \text{ (W)}$
$A_e = \frac{30\pi \vec{V}_{\text{oc}} ^2}{R_{\text{rad}} \vec{E}^i ^2} \text{ (m}^2\text{, alt. form)}$

FRIIS TRANSMISSION FORMULA

Far-field link budget P_t tx, P_r rx, range R
$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$ (W)
G_t, G_r are linear gains; $\lambda=c/f$; R in m; P_t, P_r in W.
Solve for R max range
$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$ (m)
Equivalent form using A_e recv. area form
$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$, $G_r = \frac{4\pi A_{e,r}}{\lambda^2}$
Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.
General form (off-axis) when patterns NOT peak-aligned
$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$
F_t, F_r normalized patterns at the link directions; $F=1$ when aligned.
Recipe 1. $\lambda=c/f$. 2. Convert dB gains to linear: $G=10^{G_{\text{dB}}/10}$.
3. If TX is “half-wave dipole” use $G_t=1.64$. 4. Plug into Friis.
Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}}$	$G_{\text{lin}} = 10^{G_{\text{dB}}/10}$
dB linear	dB linear
0	1
3	2
6	4
7	5
-3	0.5
-10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.
For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

POWER FLOW IDENTITIES

$P_{\text{rad}} = S_{\text{max}} \Omega_p R^2 = \frac{4\pi R^2 S_{\text{max}}}{D} \text{ (W)}$
$S_{\text{max}} = \frac{P_{\text{rad}} D}{4\pi R^2} \text{ (W/m}^2\text{)}$
Isotropic radiator: $S = P_{\text{rad}}/(4\pi R^2)$ (no D factor).

NOISE & SNR

Noise power thermal noise into matched receiver
$P_N = k_B T_{\text{sys}} B \text{ (W)}$
$k_B = 1.38 \times 10^{-23} \text{ J/K}$; T_{sys} system noise temp (K); B receiver bandwidth (Hz).
Signal-to-noise ratio
$\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs $\text{SNR} > 10$ dB for usable signal.

APERTURE ANTENNAS (horn, dish)

Directivity from aperture area uniform illumination
$D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) $\Rightarrow A_e \approx A_p$
Directivity from beamwidths rectangular beam
$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}$ (use rad)
Circular beam ($\beta_{xz}=\beta_{yz}=\beta$): $D \approx 4\pi/\beta^2$.
Half-power beamwidth uniform rect. illumination
$\beta_{xz} \approx 0.88 \frac{\lambda}{\ell_x}$ (rad) $\beta_{yz} \approx 0.88 \frac{\lambda}{\ell_y}$
Scaling: $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$. Double area $\Rightarrow D \times 2$, $\beta/\sqrt{2}$. Double f ($\lambda/2$) $\Rightarrow D \times 4$, $\beta/2$.

LINEAR ANTENNA ARRAY

Setup N identical elements, spacing d along axis
element pos $z_i = id$, $i = 0, \dots, N-1$; $A_i = a_i e^{j\psi_i}$
Pattern multiplication
$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$
Array factor $F_a(\theta) = \left \sum_{i=0}^{N-1} A_i e^{j k i d \cos \theta} \right ^2$
Uniform $A_i=1$, equal phase closed form
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$, $\gamma = kd \cos \theta$
Normalize: F_a/N^2 . Peak N^2 at $\gamma=0$ ($\cos \theta=0$, broad-side).
HPBW (broadside, uniform) array length $L=(N-1)d$
$\beta \approx 0.88 \frac{\lambda}{L}$ (rad)

SYMBOL REFERENCE

l — antenna length (m); $\lambda=c/f$ (m); $c=3 \times 10^8$ m/s
 I_0 — peak current (A); $\eta_0=120\pi \approx 377\ \Omega$
 $k=2\pi/\lambda$ wavenumber (rad/m); R obs. dist (m)
 $S(R, \theta)$ — power density (W/m²)
 $F(\theta)=S/S_{\text{max}}$ — normalized pattern (unitless)
 β — half-power beamwidth (rad or deg)
 Ω_p — pattern solid angle (sr)
 $D=4\pi/\Omega_p$ — directivity (unitless); ξ rad. eff.; $G=\xi D$
 $A_e=\lambda^2 D/(4\pi)$ — effective area (m²); $A_p=ld$ phys. (m²)
 R_{rad} — radiation resistance (Ω); R_{loss} loss res. (Ω)
 P_t, P_r — TX / RX power (W); G_t, G_r linear gains